CH303 : PROCESS

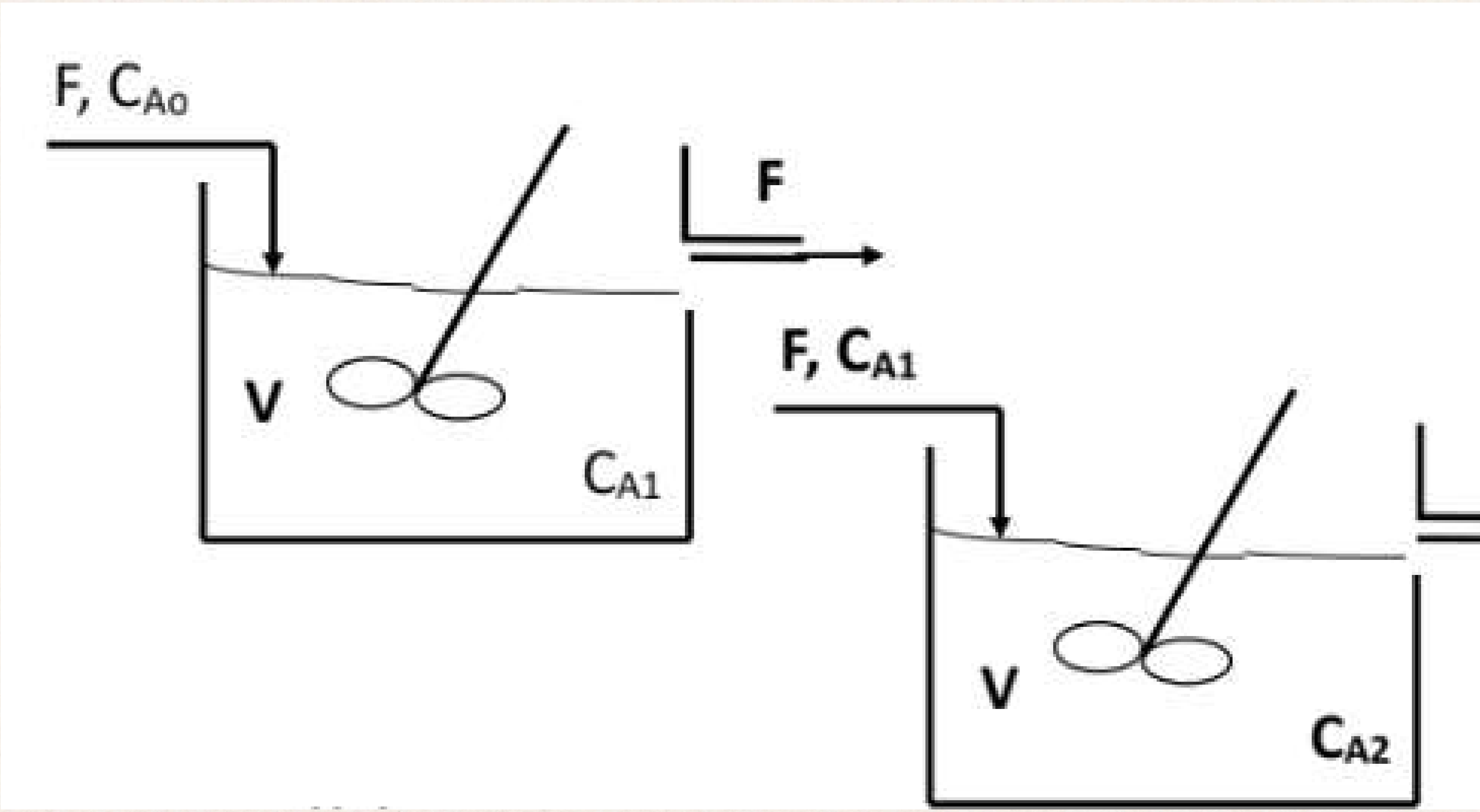
PROJECT TOPIC- CST

Kartik

ENTRY NO-2023CH

SUBMITTED TO - DR. JAYR

PROCESS DIAGRAM:



DYNAMIC MODEL EQU

COMPONENT MATERIAL BALANCE:

$$\frac{dC_{A1}}{dt} =$$

$$\frac{dC_{A2}}{dt} =$$

Given Process parameters :

CA1(initial guess) = 0.236 Mol/m3

CA2(Initial guess) = 0.05 Mol.m3

CA3(Inlet Concentration) = 0.925 Mol/m3

V(volume of

F (Feed Flow

K = 0.040 mi

Process Obje

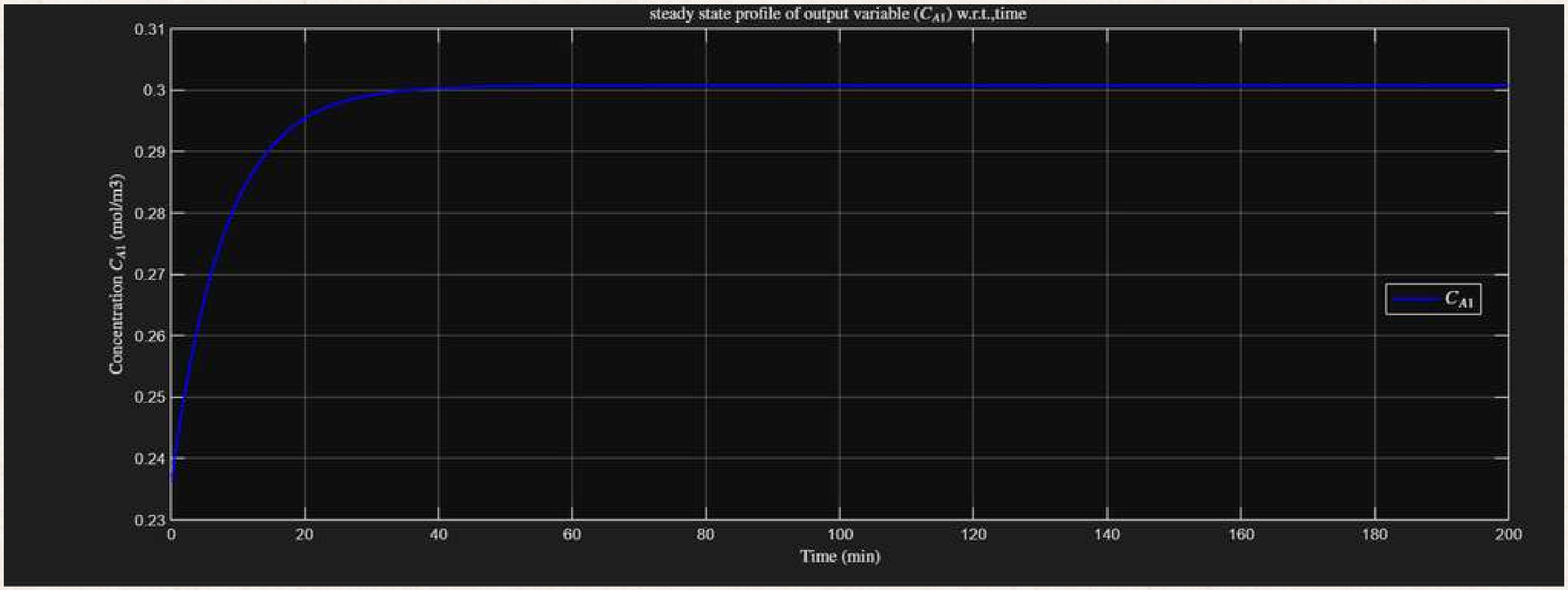
3. **Empirical Model Identification of FOPTD Models.**

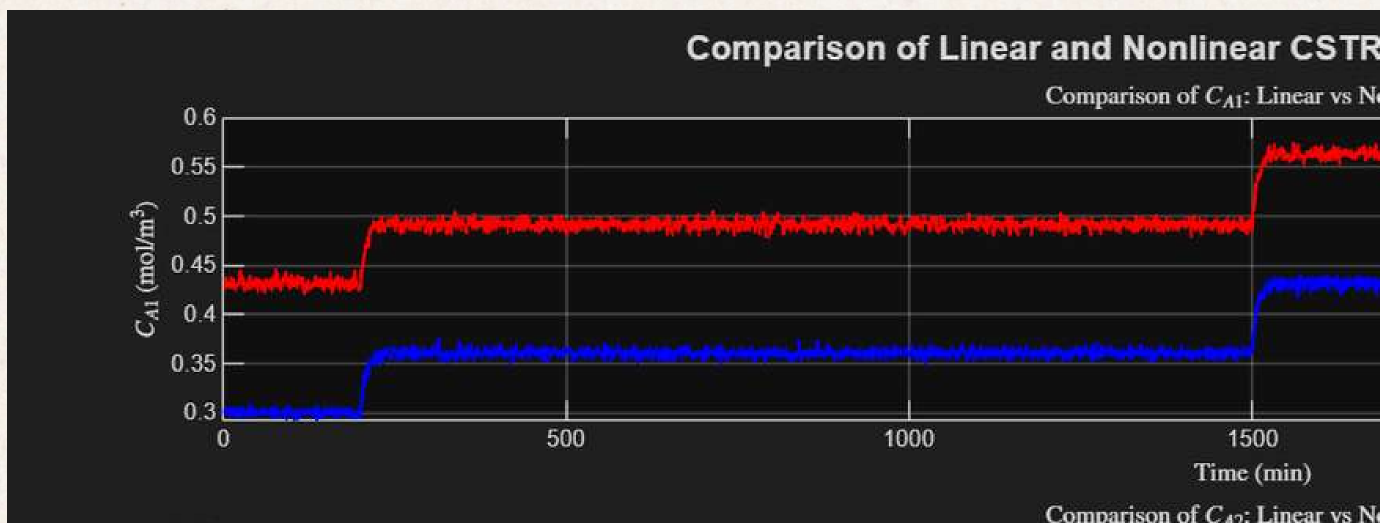
4. **Feedback Controller design**

5. **Closed loop**

OPEN LOOP SIMU

SIMULATION RESULTS OF LINEAR





Linearized Model and Trar

Linearizing the system around a steady-state allows us to use standar
and controller design.

Jacobian Linearisation:

- The nonlinear model is linearized around the steady-state operating point.
- **Jacobian matrices** are used to obtain the linearized **state-space model (A, B, C, D)**.

$$J = \begin{bmatrix} -0.1245 & 0 & 0.0405 & 0.3281 \\ 0.0405 & -0.1245 & 0 & 0.0886 \end{bmatrix}$$

Transfer Function Of C
input CA0 and F:

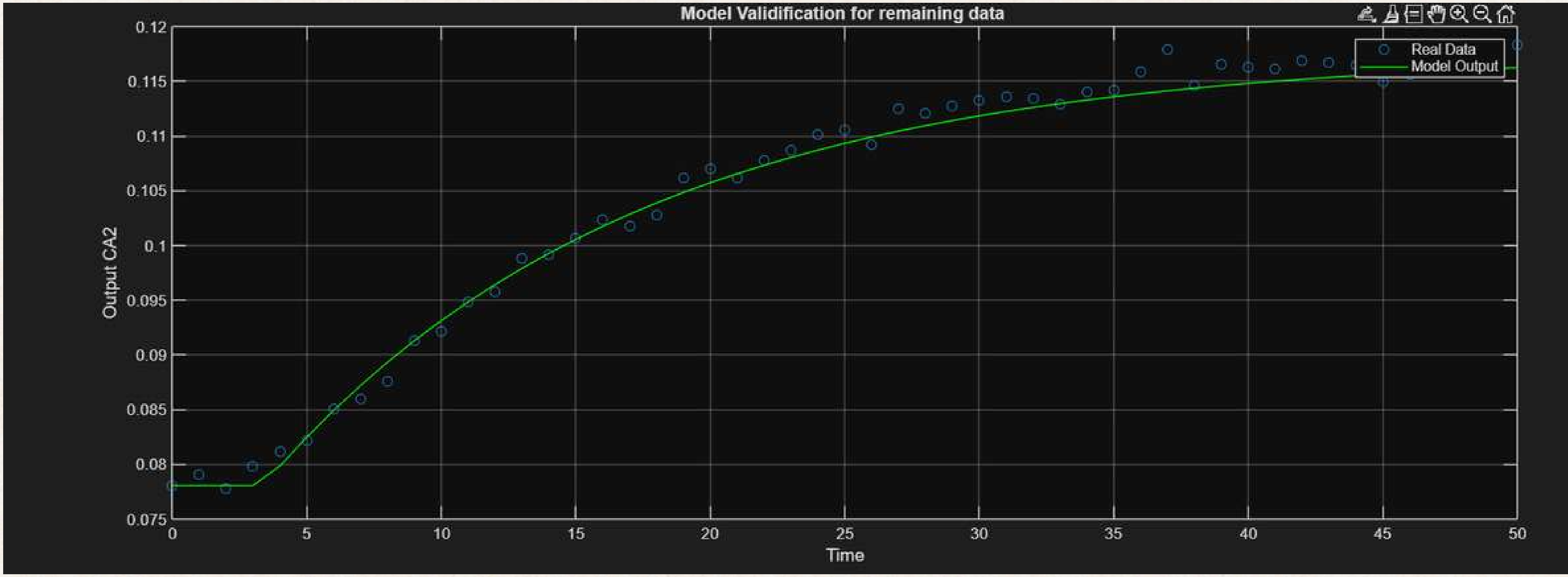
G(s)_CA1_CA0=

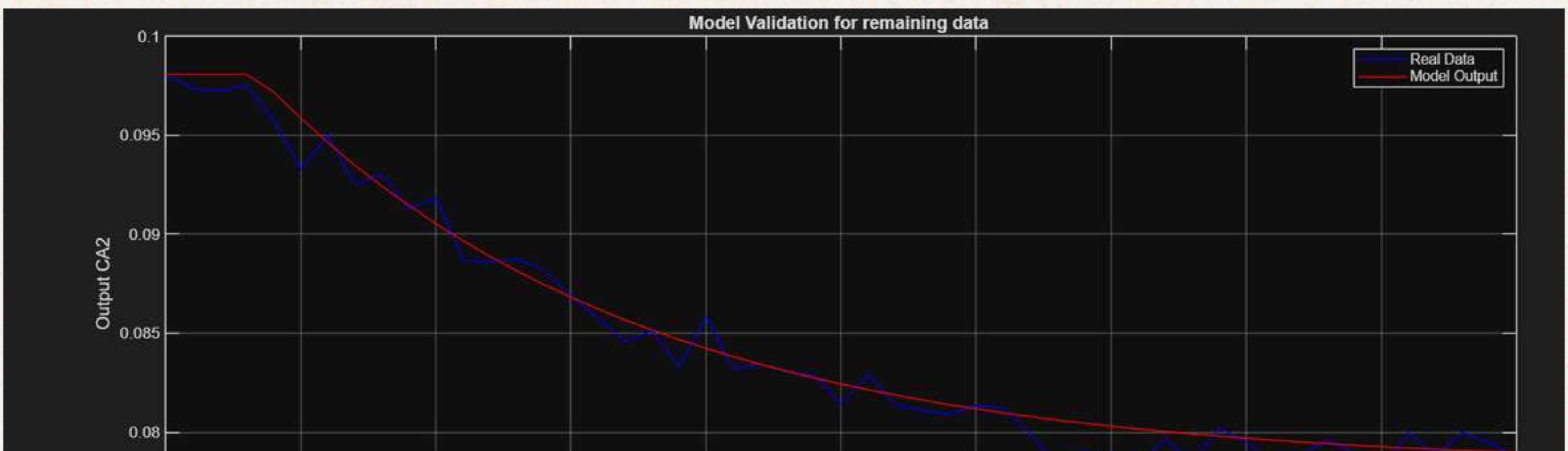
```
 0.04048
----------
s + 0.1245
```

G(s)_CA1_F =

```
 0.3281
----------
s + 0.1245
```

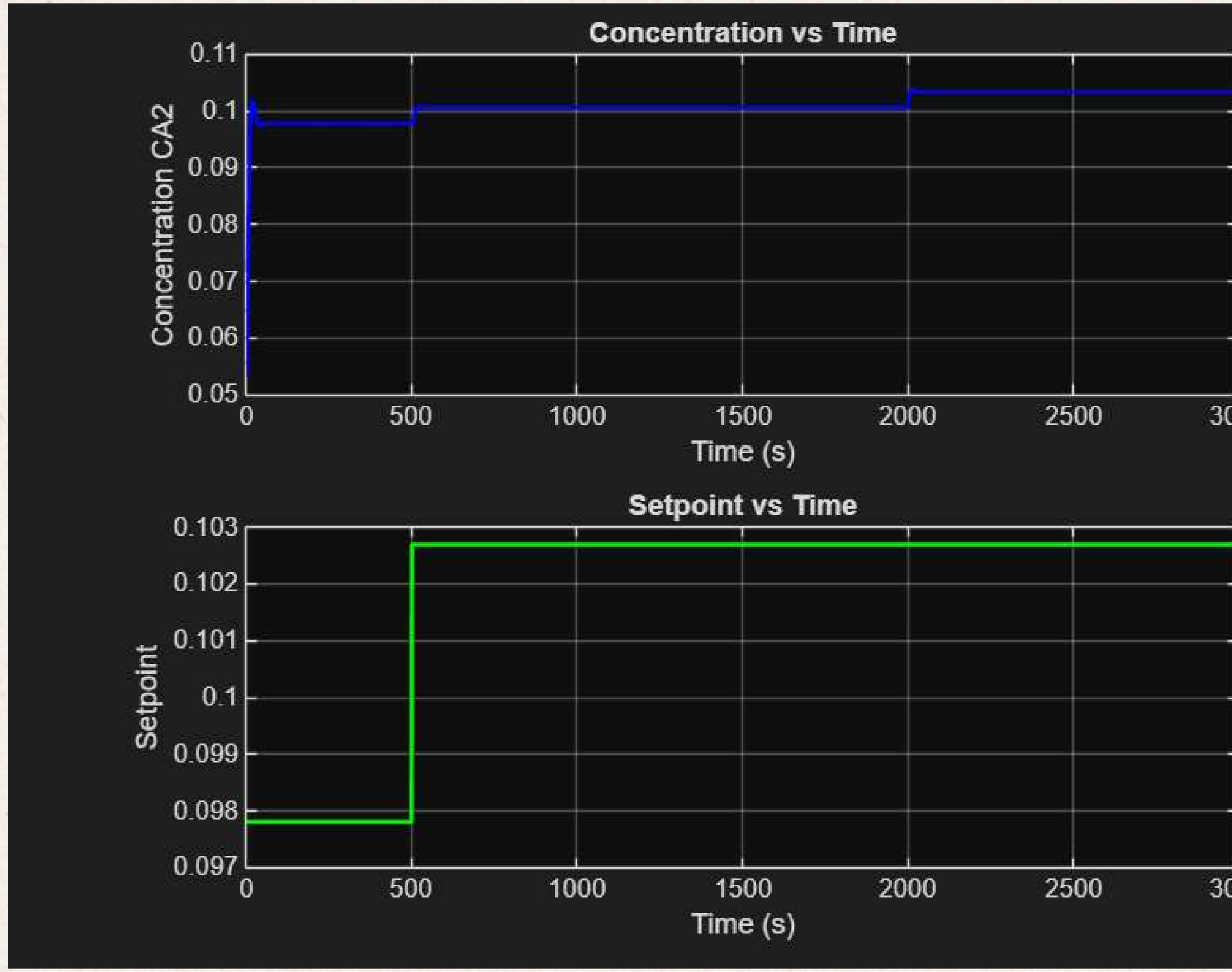
Empirical Model Identification of FOPTI





Feedback Controller design

FOR P CONTROLLER :



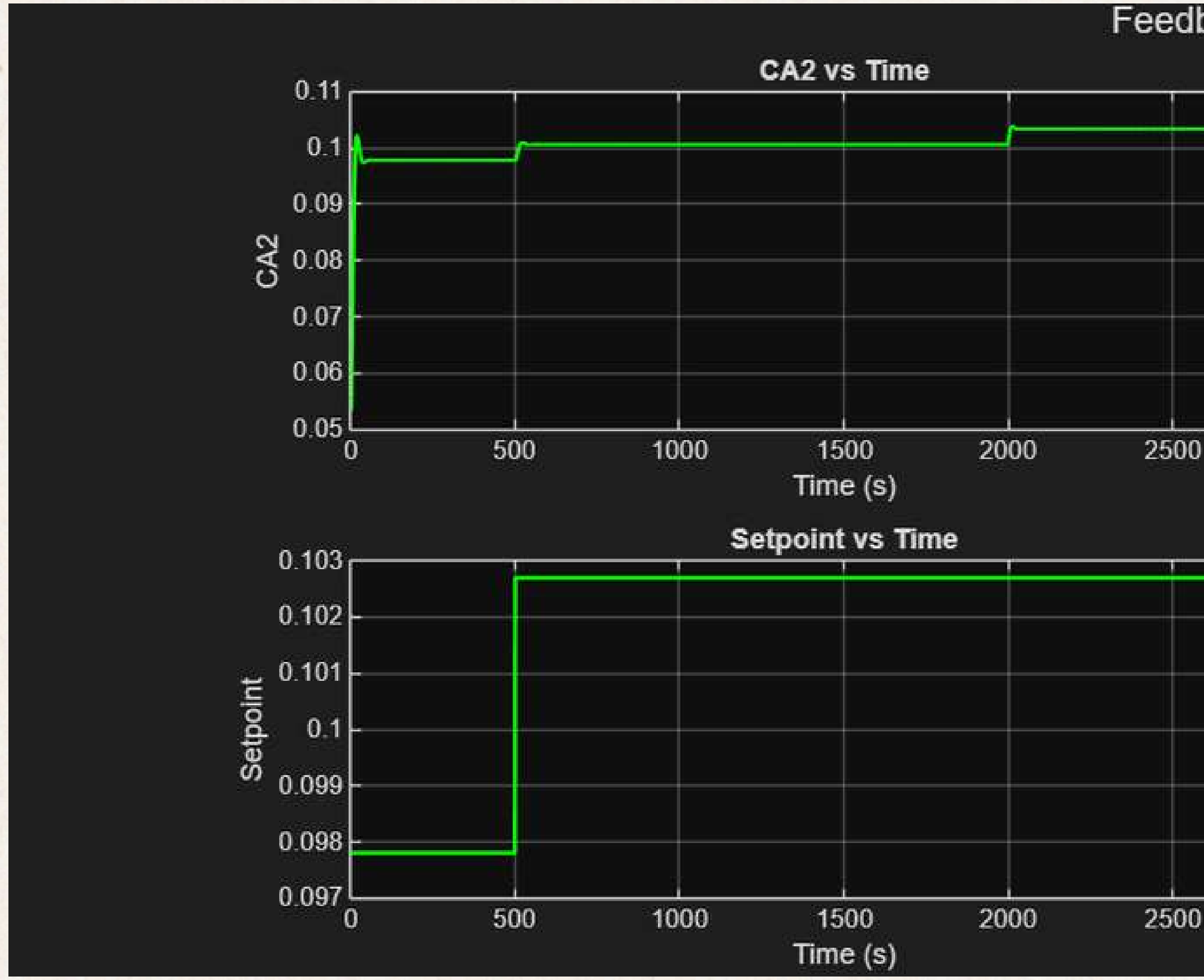
Controller Output :

P(t) = P_ss + Kc * error(t)

- **Comparison of Se controlled variab of plant w.r.t nu samples (i.e**

Feedback Controller design

FOR PI CONTROLLER :



$$u(t) = P_{ss} + K_c e(t) + K_c \frac{1}{\tau_I} \int_0^t e(\tau)\, d\tau$$

- **Comparison vs-controlled (CA2) of number of s**

CLOSED LOOP STABILITY :

Routh–Hurwitz Condition :

- **Ensures stability by checking signs of first-column.**
- **Gives stable range of Kc.**
- **All poles remain in left-alf plane**

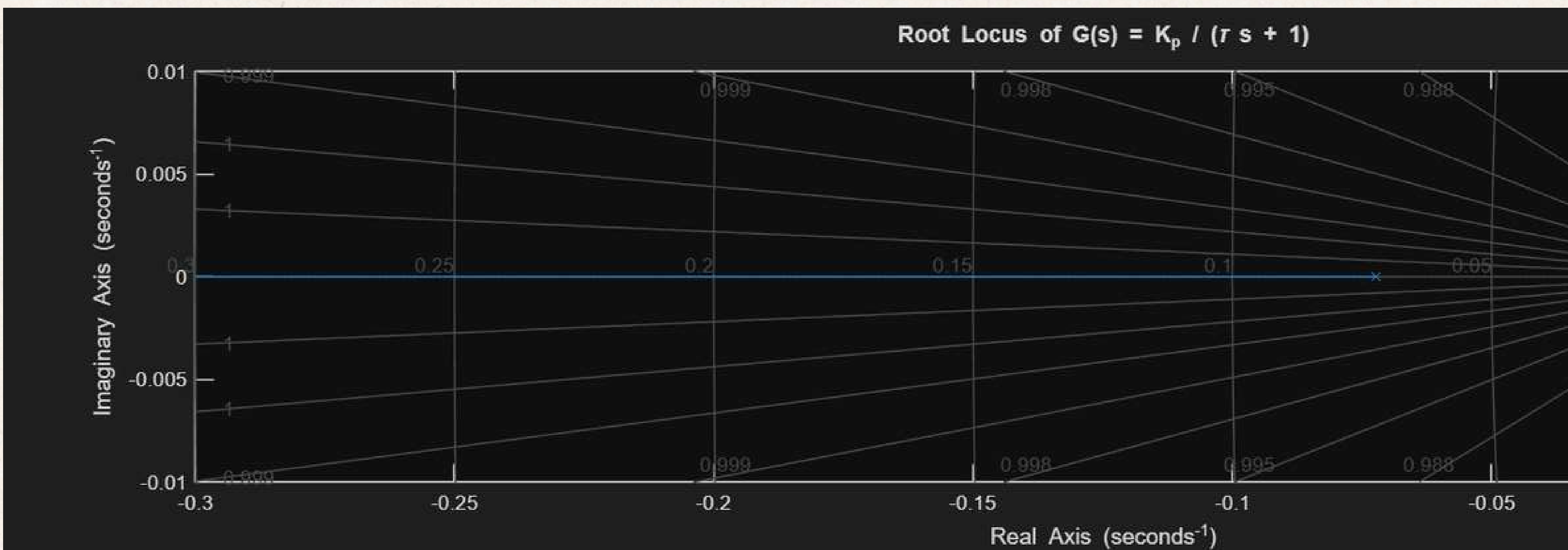
From Closed loop stability analysis , we get the conditio

1 + Kp *Kc > 0 or Kc > -

Kc should be positive for physical systems.

ROOT LOCUS DOAGRAM :

- Root Locus shows how the closed-loop poles
  as the controller gain Kc varies from 0 to ∞.
- The pole locations determine stability, speed
  oscillatory behavior.
- If poles stay in the left half-plane, the system is sta
- When poles cross into the right half-plane, th



THANK YO